



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Business Systems

Academic Year 2023– 2024 (Odd/Even Semester)

Degree, Semester & Branch: B.Tech, III & CSBS

Course Code & Title: AD3491 & Fundamentals of Data Science and Analytics

Name of the Faculty member(s): Dr. M. Gomathy Nayagam

Innovative Practice Description

- **Unit/Topic: Unit II/Theory to Practice: Correlation and Scatterplots**
- **Course Outcome: CO2**
- **Topic Learning Outcome: TLO5**
- **Activity Chosen: Theory to Practice with Handson**
- **Justification:**

Significant correlations between various measures or groupings of metrics can be found through correlation analysis. A scatterplot gives you a statistical and visual way to assess how strongly two variables are related. When assessing the strength of correlations revealed by a fishbone diagram, scatter plots can be a useful tool. In order to understand the correlation between two variables/a set using scatter plot, this activity will help the students to implement scatter plot using python and they can understand the correlation in very easy manner.

- **Time Allotted for the Activity: 50 Minutes**
- **Details of the Implementation:**
 - Course instructor explained the basic concept and he explained the usage of correlation and scatter plot in data science process with an example to the students for 30 Minutes in the class
 - After completion of the class, the course instructor gave the data set to the students and he explained the step by step implementation of finding correlation coefficient and shown the data in scatterplot using Python and Google co-lab.
 - The fig. shows the sample glimpses of hands-on session in Data Analytics laboratory
 - The students find the correlation for given dataset and segregate the relationship between each attribute in data set.

• **CO–PO/ PSOMapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 2	3	3	3	2	2	1	-	1	1	1	1	1	3	3	1

(1– Low 2–Moderate 3 – High)

• **PO/ PSOMapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5
	3	3	3	2	2
Justification for correlation	The students could be able to apply the basic mathematical knowledge to analyze the data set	The students could be able to formulate the solution for given problem using Correlation matrix and displayed the graphical result using Scatterplot	The students could be able to provide the solution for identifying the relationship between the attributes in dataset using correlation co-efficient and matrix	The students could be able to build the graphical analysis for correlation between attributes in data set.	The students could be able to use the appropriate Matplotlib, Numpy, Scipy packages to find the correlation and drawn the correlation matrix in graph

Innovative practice	PO6	PO8	PO9	PO10	PO11	PO12
	1	1	1	1	1	1
Justification for correlation	The students could be able to apply data science concept and solve the problem of societal, health related	The student could be able to know the ethical principle and responsibilities when they use public dataset	The students could be able to know their responsibility to implement concept using python as an individual.	The students could be able to communicate their results of the problem and discuss with their peer team members	The students could be able to apply the correlation matrix and find the importance of feature of dataset when they do mini project in this course	The students could be able to learn the newer package

Innovative practice	PSO1	PSO2	PSO3
	1	1	3
Justification for correlation	The students could be able to apply the Machine learning technology	The students could able to develop data science application	The student could able to apply the concept of descriptive analysis in real-time project

• **Images/Screenshot of the practice:**

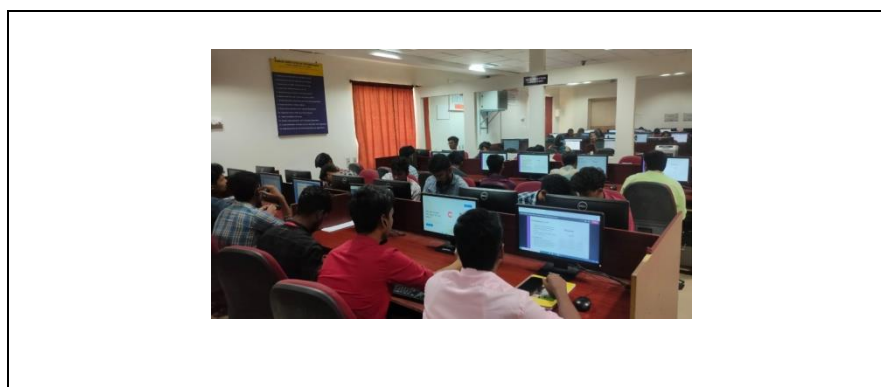


Fig.1. Glimpses of Scatter plot and Correlation Hands-On session in Data Analytics Laboratory

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

All the students are involved and worked in the hands-on session.

❖ **Benefit of the Practice (E.g. Outcome Attainment would have increased due to innovative practice over conventional practice)**

- It helps students to know the implementation of correlation coefficient for real-time dataset.

❖ **Challenges faced in implementation:**

- Make all students to implement and correct the error in the model.

References:

- ❖ Mr.M.GomathyNayagam, "OpenMP execution Model" in course CS6801 Multi-core Architecture and Programming, <https://www.ritrjpm.ac.in/images/computer-science/CS6801-MCAP-MGM.pdf>

Signature of Faculty Member
(Dr.M.GomathyNayagam)

HOD